AI Product Engineering Internship Assessment

PatentPilot – AI-assisted Freedom-to-Operate (FTO) Workspace

Submission Deadline: Thursday, 16 July 2026, 10:00 PM IST

Introduction

One important stage in the drug discovery process is evaluating whether a newly generated molecule may overlap with existing intellectual property. Before researchers invest additional time in optimization or synthesis, they often perform an initial Freedom-to-Operate (FTO) assessment to understand whether similar molecules or inventions have already been disclosed in patents.

This process is typically time-consuming and requires searching large patent databases, reviewing multiple documents, and manually preparing reports.

Your task is to design and develop a lightweight application called PatentPilot, an AI-assisted workspace that helps researchers perform an initial patentability review using publicly available patent information.

Objective

Build a web application that allows a researcher to:

- Search for patents relevant to a molecule.
- Review and organize retrieved patent information.
- Generate an AI-assisted patentability report.
- Maintain a history of previous analyses.

Data Sources

Use publicly available scientific resources wherever appropriate.

Examples include:

- SureChEMBL
- PubChem
- Google Patents (public data where applicable)

- Other publicly accessible patent or molecular databases

You are free to choose your own retrieval strategy and APIs.

Functional Requirements

1. Molecule Submission

The application should allow a researcher to submit: Molecule (SMILES) , Target (optional) , Disease or indication (optional)

2. Patent Discovery

Based on the submitted molecule, retrieve relevant patent information from public sources. The objective is **not simply to display search results**. Your application should attempt to identify the **most relevant patents** using an approach of your choice.

Possible approaches include:Molecular similarity , Keyword search , Semantic search , Hybrid retrieval, Embedding-based search

3. Patent Review Workspace

Display retrieved patents in an interface that allows researchers to efficiently review the results.

Each patent should present information such as:

- Patent title
- Patent number
- Publication date
- Assignee
- Abstract
- Source
- Similarity or relevance score (if applicable)

4. AI-assisted Analysis

For each retrieved patent, generate an AI explanation that answers questions such as:

- Why was this patent retrieved?
- Which aspects appear similar?

- What possible overlap exists?
- How confident is this assessment?

The explanation should be based on the retrieved information and should avoid generic summaries.

5. Patentability Report

Once the researcher completes the review, generate a structured report containing:

- Executive Summary
- Key Similar Patents
- Potential Novelty Concerns
- Patents Requiring Manual Review
- Overall Recommendation

The recommendation should clearly indicate one of the following:

- Low Patent Risk
- Requires Expert Review
- High Patent Risk

The report should also explain how this recommendation was reached.

You are encouraged to define and document your own scoring methodology.

Technical Requirements

You may use any programming language or framework.

Suggested technologies include:

- React / Next.js
- Spring Boot/ Node.js / FastAPI
- PostgreSQL / MongoDB
- LangChain
- OpenAI
- Gemini

- Groq
- Vector databases
- Any additional libraries you consider appropriate

Expected Deliverables

Your submission should include:

- GitHub repository with README and source code
- Architecture diagram & setup instructions

README Requirements

Your README should clearly explain:

- Overall architecture
- Retrieval strategy
- AI workflow
- Technologies used
- Assumptions made
- Trade-offs
- Future improvements
- Instructions for running the project locally